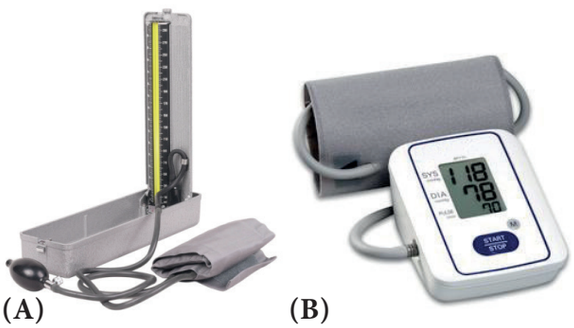
stethoscope on the chest. It is a useful diagnostic tool to identify and localize health problems and diagnose disease. The modern electronic stethoscopes are high precisioned instruments.



Figure 14.14 Stethoscope

Sphygmomanometer

Sphygmomanometer is a clinical instrument used to measure blood pressure when a person is in a relaxed and resting condition. The pressure of the brachial artery is measured. It helps to estimate the state of blood circulation and the working of the heart. It helps to diagnose conditions such as increased or decreased blood pressure. **Monometric** and **modern digital** types are the apparatus used to measure blood pressure.



(A) **(B)**

Figure 14.15 Monometric (A) and Digital (B) type blood pressure apparatus

14.17 Blood Groups

The concept of blood grouping was developed by **Karl Landsteiner** (1900). He identified blood groups **A**, **B** and **O. AB** blood group was recognized by **Decastello** and **Sturli** (1902).

Human blood contains certain specific substances called **agglutinogens** or **antigens (Ag)** and **agglutinins** or **antibodies** (**Ab**). Antigens are found on the membrane surface of **RBC**. Antibodies are present in blood plasma. Based on the presence or absence of antigen and antibodies human blood group is classified into four groups called **A, B, AB** and **O**. An individual has one of the four blood groups.

(i) 'A' group individuals: **Antigen A** is present on the surface of RBC and **antibody b (anti-b)** is present in the plasma.

(ii) 'B' group individuals: **Antigen B** is present on the surface of RBC and **antibody a (anti - a)** is present in the plasma.

(iii) 'AB' group individuals: **Antigens A and B** are present on the surface of RBC and both the **antibodies are absent** in the plasma.

(iv) 'O' group individuals: **Antigen A or B** are absent on the surface of RBC. However, the plasma **contains both the antibodies** a and b **(anti a** and **b)**.

Blood donation

In blood transfusion one must consider the antigen and antibody compatibility (matching) between the donor and the person receiving blood (recipient). When an individual receives a mismatched blood group from the donor **agglutination (clumping)** of blood occurs in the body which leads to death.

Persons with '**AB**' blood group are called '**Universal Recipient**' as they can **receive blood** from persons with any blood group.

Persons with '**O**' blood group are called '**Universal Donor**' as they can **donate blood** to persons with any blood group.

Rh factor

Rh factor was discovered by **Landsteiner** and **Wiener** in 1940 in **Rhesus monkey.** The surface of RBC contains the antigen for Rh factor. **Rh^+** (positive) persons have Rh antigen on the surface of RBC while, **Rh^-** (negative)

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